

Response Letter with Annotations

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Annotations

1. Purposes

a. Motivation (one-sentence summary)

Few to no prior studies attempt to systematically map the kinds of information one could see on social media platforms and compare that pool of information to what users actually see.

b. Specific Rationale (several bullet points)

- Their study is an update from previous work that is outdated (from 2014) that doesn't examine the impact of content posted in groups and by pages
 - This is especially important because people may select groups and pages more by ideological concordance than friends
- And other previous studies have sampling issues that limit generalizability (convenience, opt-in sampling)
- Domains/sources are more complex than simply by site (i.e., Wall Street Journal newsroom is very different than their very conservative leaning opinion section)

c. Research Questions and/or Hypotheses

- Identify IV, DV, mediator, moderator
- Identify specific relationships among variables

The authors examine “supply and demand” for political news by tracking a few things:

- Inventory of all content users could have seen
- Subset of content that users actually ended up seeing
- And who engages with what content via clicks, likes, reactions, reshares, and commenting

The authors had 6 key research questions:

RQ1: How ideologically segregated is news consumption on Facebook, and are those patterns of segregation symmetric on the right and left?

RQ2: How does segregation vary with potential news consumption versus actual exposure versus engagement?

RQ3: How does segregation vary if the level of analysis is URLs rather than domains (thus capturing curation of content within domains)?

RQ4: How segregated is exposure on Facebook relative to the benchmark of browsing behavior (the predominant source of data in past research)

RQ5: How segregated are the streams of content from the major pathways to exposure on Facebook (friends, Pages, and Groups)?

RQ6: How prevalent is exposure to unreliable content on the right relative to the left?

2. Method

a. Research Design

The authors calculated segregation index and favorability scores, then conducted a network analysis looking at the ties between domains to assess segregation of news sources, URLs to assess segregation of sources at the news-story level, and coexposure networks where nodes = domains or stories, and edges = the number of unique users exposed to both nodes.

Social media runs on algorithmic and social curation processes that feed back into the algorithm in a loop. The authors only analyzed ~3.9% of content that U.S. adults saw on the platform during the study period (posts flagged by Facebook's internal algorithm as "political news" and contained a URL).

b. Sample (representative sample?)

Sample was representative of all U.S. Facebook users (N = 231 million)

c. Measures

- They set cutoffs for ideology scores at the poles (≤ 0.35 for liberal, ≥ 0.65 for conservative)
- Used a political segregation index adapted from research on residential segregation by political partisanship
- Used favorability scores associated with individual domains and URLs to infer ideological composition of their audiences

d. Analysis

- Authors calculated the segregation index and favorability scores for their three groups of audiences: potential, exposed, and engaged
- Then conducted a network analysis to assess clustering among exposed audiences by assessing how strong these networks are without considering audience ideology

3. Results and Conclusions

a. Answers to RQs and Hypotheses

RQ1: The segregation score based on exposed audience domains fluctuates around 0.35, or a 35 percentage point gap between liberals and conservatives compared to overlap within their respective groups.

RQ2: As engagement funnels down, ideological segregation increased. Segregation scores for potential audiences are higher than previous research results, and exposed audiences are higher than potential scores in this study. Engaged audiences are the highest of all – "as we move down

the funnel of engagement, liberal and conservative audiences become more isolated from each other.”

RQ3: Evaluating by URLs raises the segregation score further between 0.45 and 0.5 (so nearly a 50% gap) compared to domains alone (like in RQ1).

RQ4: Segregation of groups was greater across all levels of engagement compared to prior research when comparing to web activity. Segregation score was three times as high on Facebook compared to news exposure on other web activity.

RQ5: Misinformation shared by pages and groups had more homogenous audiences concentrated on the far right of the ideology scale.

RQ6: Most sources of misinformation were favored by conservative audiences, and these distributions did not shift down the funnel of engagement like segregation scores did.

b. Include Key Figures/Tables

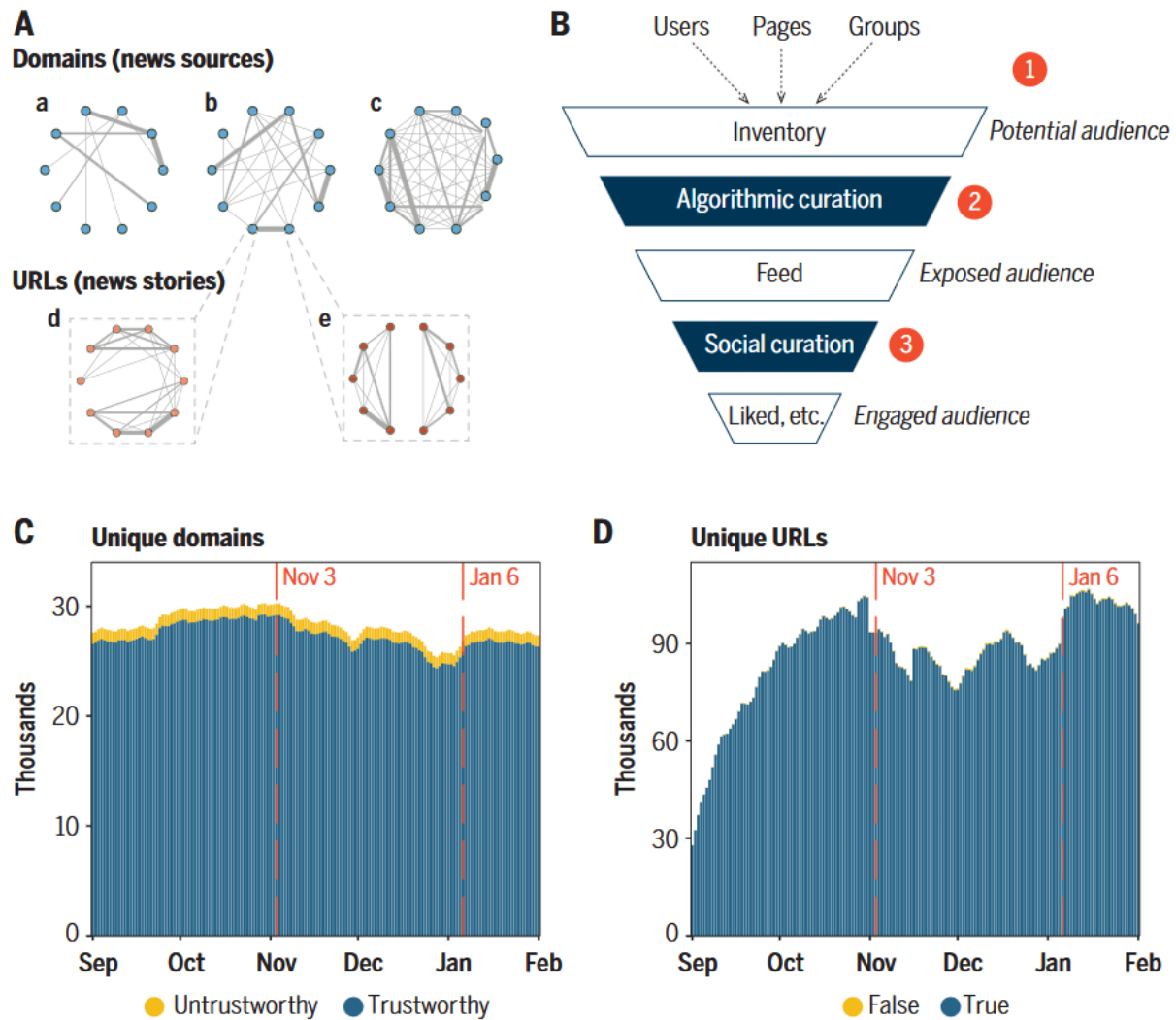


Fig. 1. Summary of data and levels of analysis. (A) Our analyses center on three different levels: domains (to calculate segregation metrics at the news source level), URLs (to calculate segregation metrics at the news-story level), and coexposure networks, where nodes are domains or news stories and the edges encode the number of unique users exposed to a given pair. The analysis of coexposure networks allows us to determine whether there is notable clustering in exposure to sources and, within sources, in exposure to stories. The schematic examples in subpanels (a) to (c), for instance, do not show strong evidence of clustering, but the networks in subpanels (d) and (e) do. (B) Schematic representation of the funnel of engagement (from potential exposure to actual exposure to engagement). Numbers indicate the level of analysis implied in our RQ2: (i) Are news inventories ideologically segregated, (ii) does segregation increase after algorithmic curation, and (iii) does segregation increase based on what content users choose to engage with. (C and D) Number of news domains (categorized as untrustworthy or trustworthy) (C) and news URLs (rated false and not rated false by Meta's 3PFC) (D) in our data, presented as daily counts. The increase in the number of unique news stories depicted in (D) between September and November is partly an artifact of the threshold that we impose on the analysis of URLs (for privacy reasons, we only analyze URLs that were shared more than 100 times by US-based users during the observation period). Yellow indicates the volume of news domains or news URLs that were rated untrustworthy or false, and blue indicates the volume of news domains or news URLs that were not rated untrustworthy or false. The red dashed lines highlight November 2 and January 6, which are the dates of the election and the US Capitol riot, respectively.

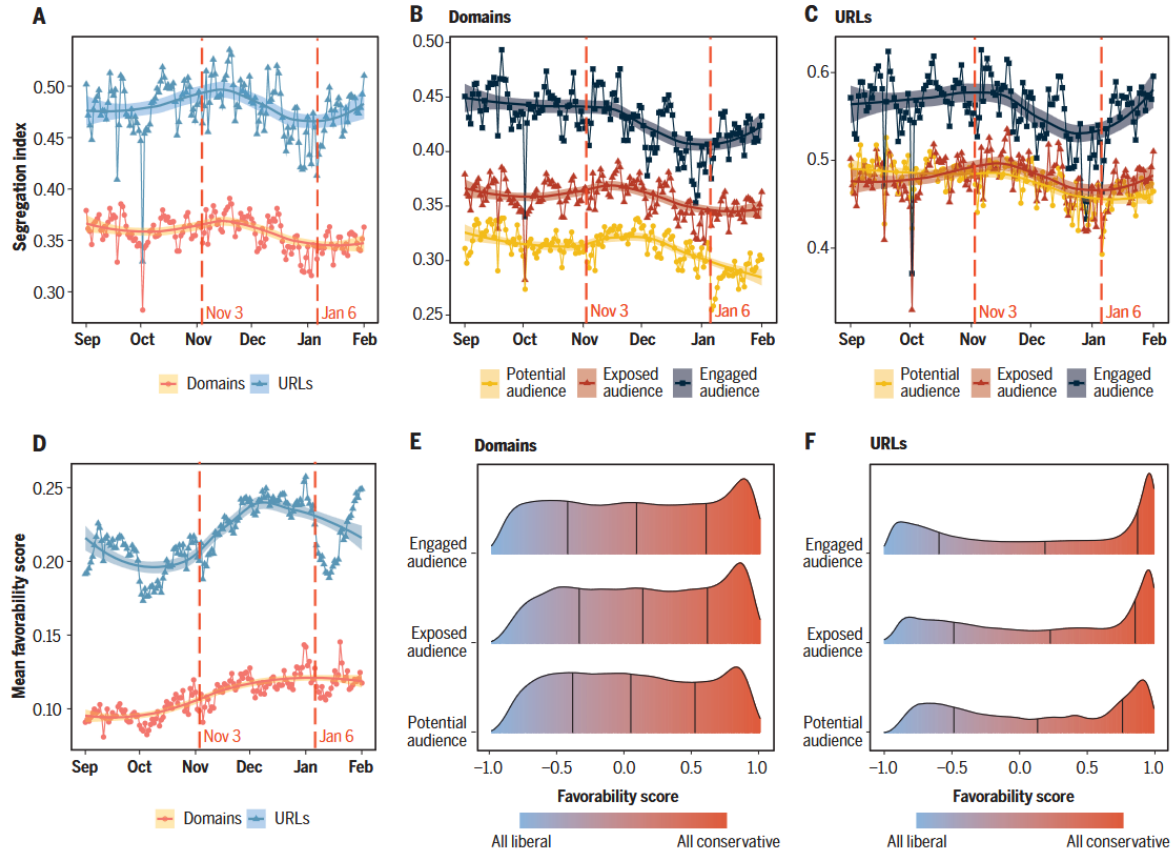


Fig. 2. Segregation and audience polarization at the domain and URL levels.

(A) The segregation score based on exposed audience and calculated according to Eq. 1 is consistently higher at the URL level, suggesting that there are information curation practices with news stories that get masked when aggregating the data at the domain level. (B and C) Segregation scores drawn from exposed audiences are higher than those based on potential audiences but lower than the scores from engaged audiences (the difference between potential and engaged audiences is only visible at the domain level). This suggests that algorithmic and social amplification both contribute to segregation (note that we use different y-axis ranges to facilitate

comparison across curves for domains and URLs). (D) The mean favorability scores (calculated according to Eq. 2, with -1 indicating a homogeneously liberal audience and 1 a homogeneously conservative audience) suggest that, overall, audiences that are consuming political news have a right-leaning slant (i.e., all scores are above the zero line). (E and F) The density plots show that the distribution is substantially skewed toward the right, with more domains and URLs being favored by very conservative audiences. Vertical black lines mark the quantiles of the distributions; scores are calculated for each domain and URL according to Eq. 2. For (A) to (D), shaded regions indicate 95% confidence intervals for the time trend based on a local polynomial regression.

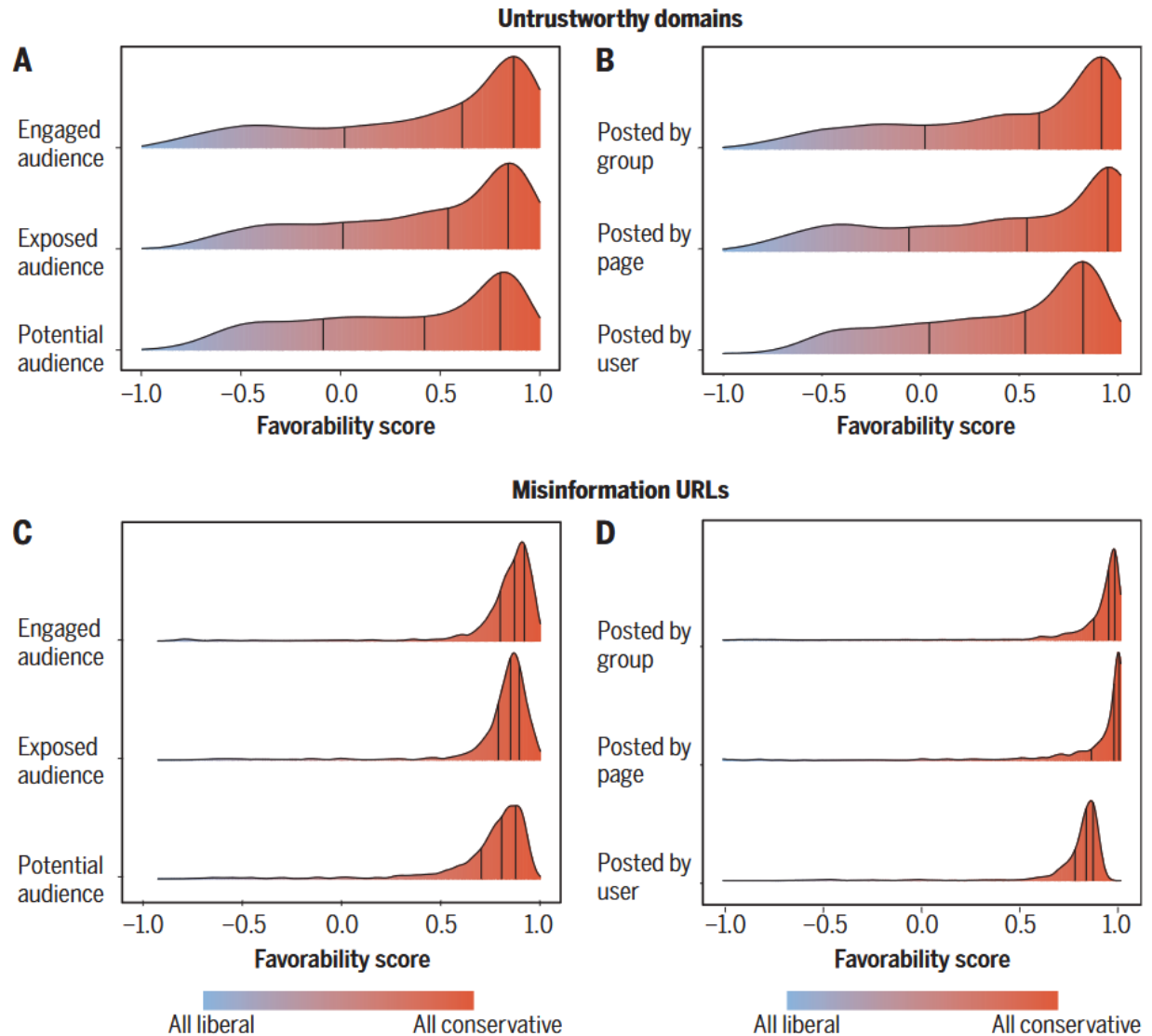


Fig. 3. Audiences of untrustworthy domains and false URLs. (A) The density plots show that most domains that are categorized “untrustworthy” are favored by audiences that are predominantly conservative. For potential, exposed, and engaged audiences, 71, 76, and 76% of untrustworthy domains have audiences that are conservative on average (favorability score > 0). (B) The audience of domains posted by Pages is the most conservative. (C) The data suggest a stark conservative-leaning slant for news stories labeled as “misinformation.” For potential, exposed, and engaged audiences, 97% of false URLs in each case have audiences that are conservative on average. (D) The sharing of news stories labeled as “false” is much more concentrated among conservative Pages than among Groups and users. Vertical black lines mark the quantiles of the distributions; scores are calculated for each domain and URL according to Eq. 2.

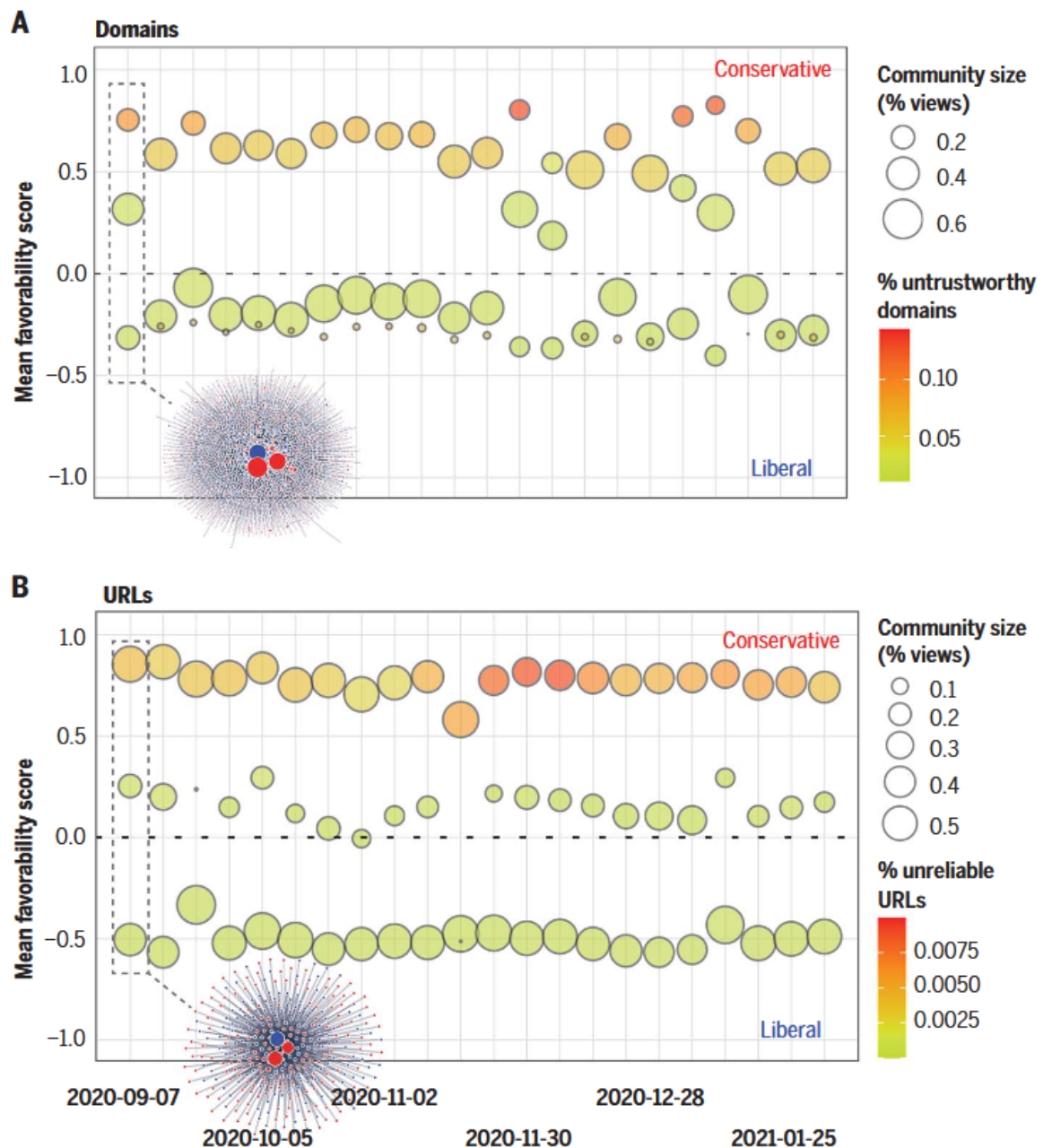


Fig. 4. Top communities in coexposure networks. (A and B) Mean favorability scores for the top three communities by views in the weekly coexposure networks for domains (A) and URLs (B). The graphic in the lower-left part of each panel shows the full network for week 1 collapsed to the communities (i.e., each node is a group of domains and URLs with higher audience overlap). The presence of clusters in these networks suggests selective exposure: The networks are clearly organized around large clusters formed by mostly liberal or mostly conservative audiences, regardless of the level of aggregation (domain or URL). These clusters also provide additional evidence of asymmetric ideological segregation, which is especially visible in the URL networks: Across the observation window, there is a sizable cluster of news stories consumed predominantly by very conservative audiences who are exposed to a higher percentage of unreliable news stories (color coded).

c. Authors' Conclusions

The authors concluded that the ideological segregation on Facebook is substantial, and especially so on the conservative side of the ideology scale.

4. Discussion

a. Implications

The authors state that Facebook is very segregated ideologically, and far more so than previous research has found on other Internet settings, media, and browsing behavior. Individual preferences and the platform's algorithm are likely playing in a feedback loop – because of the prevalence of the content posted by pages and in groups where these ideological segregations are more apparent. People choose the kinds of groups and pages they want to be in which feed into the algorithm more posts that align with their ideological stances.

b. Limitations

- The authors acknowledge the limitation that they only looked at domains and URLs and didn't include other kinds of platform content that could have different patterns altogether.
- The authors relied on Facebook's classifiers for these domains and URLs because they "perform better than ad hoc alternatives" and state that testing the robustness of their results requires more research

c. Future Directions

The authors state that future studies should incorporate non-URL content and expand the kinds of coverage/topics that misinformation can occur in, using human annotation or automated classification methods.

Response Letter (~600 words)

- Your assessment of the study

Overall, this study provides great (and concerning) insight into how these webs of ideological segregation have formed on Facebook and how the platform has contributed to political polarization in U.S. society (the main goal of the FIES project). I think their assessment of the feedback loop of the user-informed algorithm creating a spiral of ideological segregation is accurate and supported by their study's findings. Though the study couldn't account for everything, it builds upon prior work that was long outdated and no longer describes the Facebook platform accurately compared to its existence 6(?) years prior.

- Weaknesses

I became concerned about the study's methods when I read that the authors used Facebook's internal metric for deciding what is political news or not, and then even more so when they only included posts that contained URLs. The authors state, "in other words, the algorithmic curation process that we analyzed here relies on many more signals than those generated by the content we analyzed (i.e., political domains and URLs)."

Many political posts and especially rage bait, share bait, and other generally misinformative posts come in the form of *images* – and sometimes that image is a screenshot of a news source with some charged political commentary on it! Not to mention the amount of "candid" video rants from random "influencers" that exist now too that never cite sources or give any coherent reasoning for their opinions. Why would the authors not implement some third-party assessment of political content, or attempt to account for images posted by groups and pages sorted by ideology? Maybe another FIES paper covered this instead. The authors did acknowledge that this is a shortcoming, and I don't think it necessarily makes their results any less unsettling. Perhaps we were also methodologically catching up to being able to identify rage and share bait images on a large scale while this study was being designed.